A Project Report on

Krushi-Yantra: Gamified Climate-Safe Farming with AI & Carbon Income

Submitted in partial fulfilment of the requirements
of the degree of

BACHELOR OF ENGINEERING

By

Sarvesh Bhartu: 23106055

Rajanya Kshatriya: 23106096

Ramnarayan Maurya: 23106119

Sagar Mane: 23106032

Under the Guidance of

Dr. Tejashri Kolhe

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

(2025-2026)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled **"Krushi Yantra: Gamified Climate-Safe Farming with Ai and Carbon Income"** is a bonafide work of **Sarvesh Bhartu (23106055), Rajanya Kshatriya (23106096), Ramnarayan Maurya (23106119), Sagar Mane (23106032)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

Dr. Tejashri Kolhe Project Guide	Prof. Yogeshwari Hardas Project Co-Ordinator
Dr. Jaya Gupta Head of Department	Dr. Uttam D Kolekar Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for T. E.

This project report entitled **Krushi Yantra: Gamified Climate-Safe Farming with Ai and Carbon Income** by ***Sarvesh Bhartu, Rajanya Kshatriya, Ramnarayan Maurya Sagar Mane*** is approved for the degree of ***Bachelor of Engineering*** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**, ***2025-26***.

Examiner Name	Signature
1. _____	_____
2. _____	_____

Date:

Place:

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Sarvesh Bhartu
23106055

Rajanya Kshatriya
23106096

Ramnarayan Maurya
23106119

Sagar Mane
23106032

Date:

Abstract

Agricultural productivity and food security in India remain constrained by three persistent gaps: rural smallholder farmers lack localised, data-driven crop guidance and early-warning systems for invasive species; and government administrators have no intelligent tool to monitor farm submissions and derive actionable policy insights from platform data. Krushi-Yantra is a full-stack, AI-driven platform that addresses both dimensions simultaneously.

Rural farmers can map their land using a multi-zone polygon tool on satellite imagery, identifying crop areas separately from houses, water bodies and storage. Official government land documents (7/12 Extract, Khasra, Patta) are verified via AI-powered OCR and cross-matched against mapped zones to prevent land fraud. Carbon credits are calculated exclusively on verified farming zones. Farmers receive soil-aware and budget-sensitive crop recommendations grounded in Neo4j graph data from neighbouring farms, scan plants using a dual-engine pipeline combining YOLOv8 object detection with Google Gemini Vision, and access personalised advice via a Hindi voice interface without requiring internet connectivity.

Government administrators (Mayors, Panchayat heads) interact with a LangChain-orchestrated multi-agent AI system that visualises live platform data through natural-language queries. The Admin module provides territory heatmaps, disease outbreak maps, collective carbon credit pooling, and a tree-planting simulation tool with 360-degree visualisation. The platform aligns with SDG 2 (Zero Hunger), SDG 13 (Climate Action), and SDG 15 (Life on Land) by empowering farmers with data-driven intelligence and equipping governments with tools for proactive environmental governance.

Keywords: AI-Powered Agriculture; Krushi-Yantra; Precision Farming; Neo4j Graph Database; YOLOv8; Google Gemini; LangChain; Carbon Credits; Carbon Credit Marketplace; Hindi Voice Interface; Government Intelligence Dashboard.

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ABBREVIATIONS

AI	*Artificial Intelligence*
API	*Application Programming Interface*
CORS	*Cross-Origin Resource Sharing*
CSV	*Comma-Separated Values*
GIS	*Geographic Information System*
GPS	*Global Positioning System*
HTTP	*Hypertext Transfer Protocol*
JSON	*JavaScript Object Notation*
JWT	*JSON Web Token*
LLM	*Large Language Model*
ML	*Machine Learning*
NDVI	*Normalised Difference Vegetation Index*
NLP	*Natural Language Processing*
OCR	*Optical Character Recognition*
OTP	*One Time Password*
REST	*Representational State Transfer*
SDG	*Sustainable Development Goal*
STT	*Speech-to-Text*
TTS	*Text-to-Speech*
UI	*User Interface*
UUID	*Universally Unique Identifier*
YOLO	*You Only Look Once*

Chapter 1
Introduction

1.1 Background

The agricultural sector in India is the backbone of the national economy, employing approximately 42 percent of the workforce and contributing nearly 17 percent of the GDP. Yet, despite its scale, Indian agriculture remains one of the most under-digitised sectors. The majority of smallholder farmers — those owning less than two hectares of land — make critical decisions about crop selection, irrigation, and pest management based on inherited knowledge and seasonal observations. While this traditional wisdom has served farming communities for centuries, it is increasingly inadequate in the face of volatile climate patterns, soil degradation, and rapidly changing market conditions.

The first wave of agricultural modernisation — the Green Revolution — introduced high-yield crop varieties and chemical fertilisers, dramatically improving output. However, its uniform, input-intensive approach created environmental externalities including groundwater depletion, soil acidification, and biodiversity loss. The subsequent emergence of Precision Agriculture introduced the concept of site-specific crop management, using sensors and geographic information systems to tailor resource application to localised conditions. AgriAssist builds upon this foundation and takes it further by integrating Artificial Intelligence, graph-based knowledge representation, satellite imagery, and government administrative tools into a single unified platform.

A fundamental challenge in modern agriculture is not the absence of data but the fragmented and inaccessible nature of available information. Soil chemistry data exists in government databases but is not connected to individual farmer profiles. Weather forecast services provide accurate predictions but not in formats accessible to rural farmers with limited digital literacy. Plant disease libraries contain thousands of diagnostic records but require expert agronomists to interpret. AgriAssist acts as the integration layer that transforms these disparate data sources into actionable, personalised, and accessible intelligence.

The application of deep learning to plant pathology represents one of the most transformative developments in modern AgTech. Convolutional neural networks trained on large labelled datasets such as PlantVillage can now identify hundreds of distinct disease markers from a smartphone photograph with accuracy matching or exceeding trained agronomists. By embedding this capability into a mobile-accessible platform and combining it with the contextual reasoning power of large language models, AgriAssist delivers expert-level botanical diagnosis to every farmer regardless of their location or educational background.

1.2 Motivation

The primary motivation for AgriAssist emerged from three converging observations about the state of agricultural technology in India. First, existing AgTech solutions are predominantly designed for large-scale commercial farming operations with high digital infrastructure requirements. They assume reliable internet connectivity, capable hardware, and technically literate users — assumptions that are fundamentally incompatible with the reality of most Indian smallholder farmers.

Second, despite a growing global carbon credit market worth billions of dollars, Indian farmers remain almost entirely excluded from it. Yet no accessible, trusted mechanism exists to measure this contribution, verify it against land records, and connect farmers to industrial carbon buyers. AgriAssist addresses this gap by building a complete carbon credit pipeline from farm-level verification to government-mediated bulk trading.

Third, government administrators responsible for agricultural territories — Mayors, District Collectors, Panchayat Sarpanchs — lack intelligent tools to monitor the farms under their jurisdiction, visualise disease outbreaks, identify heat-stressed zones, or derive policy insights from aggregated platform data. AgriAssist introduces a LangChain-orchestrated multi-agent AI system that enables natural language interaction with live agricultural data, empowering evidence-based governance.

1.3 Aim and Scope of the Project

The primary aim of AgriAssist is to develop a production-ready, AI-powered agricultural intelligence platform that serves rural farmers and government administrators through a shared data infrastructure. The platform provides the following core capabilities:

- Zone-based satellite land mapping with official document verification and fraud prevention
- AI-driven crop recommendation using graph traversal of neighbouring farm data
- Dual-engine plant disease detection combining YOLOv8 object detection with Gemini Vision analysis
- 10-kilometre disease alert broadcasting to farms in proximity of a confirmed outbreak
- Hindi voice interface enabling farmers without internet or digital literacy to access the system via phone calls
- Territory heatmap visualisation with tree-planting simulation and temperature impact modelling
- Collective carbon credit management enabling Admins to pool, verify and trade farmer credits in bulk

The platform is scoped for deployment in Maharashtra, India, with the technical architecture designed for national scalability.

Chapter 2

Literature Survey

This chapter reviews the existing body of research and systems across six domains that form the technical and conceptual foundations of Krushi-Yantra: AI-driven agricultural decision systems, knowledge graph applications in agriculture, computer vision for plant disease detection, voice interfaces in rural technology, carbon credit mechanisms in agriculture, and government technology for territory management.

2.1 AI-Driven Agricultural Decision Systems

Wolfert et al. [1] conducted a comprehensive review of big data applications in smart farming, demonstrating that integrating heterogeneous data streams from sensors, satellites and market data significantly improves crop management outcomes. Their work established the theoretical basis for data fusion architectures in precision agriculture.

Liakos et al. [2] surveyed machine learning applications across soil management, crop yield prediction, weed and disease detection, and water management, concluding that neural network approaches consistently outperform rule-based expert systems across all agricultural domains.

Kamilaris and Prenafeta-Boldu [3] specifically examined deep learning in agriculture, reviewing 40 studies and finding that convolutional neural networks achieve mean accuracy

above 90 percent for crop disease identification when trained on sufficiently large and diverse datasets. Their work directly motivates the YOLOv8-based disease detection component of Krushi-Yantra.

Vinuesa et al. [4] analysed the potential of AI to accelerate progress across all 17 SDGs, finding that AI could enable 79 percent of the targets across SDG 2, 13 and 15 — the three goals most directly addressed by this platform.

2.2 Knowledge Graphs in Agriculture

Neo4j Inc. [5] documented the application of graph databases to connected agricultural data, demonstrating that graph traversal queries for farm neighbourhood analysis and disease propagation modelling are 40-60 times faster than equivalent relational database queries. Their architecture directly influenced the Neo4j graph schema design in Krushi-Yantra.

Deng et al. [6] proposed an agricultural knowledge graph framework integrating crop entities, soil conditions, and environmental relationships, showing that graph-based recommendation systems outperform collaborative filtering approaches for crop selection by 23 percent in precision metrics.

Pan et al. [7] demonstrated that knowledge graph embeddings can be used to predict soil-crop compatibility with an F1 score of 0.87, supporting the use of Neo4j graph traversal as the primary mechanism for crop recommendation in Krushi-Yantra. Their work also showed that incorporating neighbour farm data through graph edges significantly improves recommendation quality compared to single-farm analysis.

2.3 Computer Vision for Plant Disease Detection

Mohanty et al. [8] introduced the PlantVillage dataset — 54,000 images spanning 26 diseases across 14 crop species — and demonstrated that deep neural networks could achieve 99.35 percent accuracy in controlled laboratory conditions. Their dataset forms the training foundation for the YOLOv8 model used in the Krushi-Yantra disease scanner.

Hughes and Salathé [9] extended this work to field conditions, finding that real-world accuracy drops to 31 percent with generic models and proposed transfer learning with domain-specific fine-tuning as the solution — a technique adopted in Krushi-Yantra.

Redmon et al. [10] introduced the YOLO (You Only Look Once) real-time object detection framework, which achieves detection speeds 1000 times faster than region-proposal-based networks while maintaining competitive accuracy. The YOLOv8 iteration used in Krushi-Yantra builds on this foundation with improved small-object detection capability, which is essential for detecting early-stage leaf lesions.

Google [11] documented the Gemini multimodal model's capability in botanical analysis, demonstrating its ability to provide detailed contextual descriptions of plant conditions from bounding box crops — supporting the dual-engine detection pipeline in Krushi-yantra.

2.4 Voice Interfaces in Rural Technology

Plauché et al. [12] studied spoken dialogue systems for low-literacy users in South Africa and India, finding that voice interfaces reduce task completion errors by 67 percent compared to screen-based interfaces for users with limited formal education. Their findings directly motivated the Hindi voice interface design in Krushi-Yantra.

Parikh et al. [13] conducted user studies on mobile agricultural information services in rural Gujarat, concluding that audio-based information delivery achieves 4.2 times greater adoption among low-literacy farmers than text-based alternatives.

Twilio's documentation [14] outlines the architecture for programmable voice applications supporting speech-to-text in 119 languages including Hindi, enabling the construction of inbound call handlers that parse natural language queries and route them to backend services. The offline caching mechanism in Krushi-Yantra extends this by pre-generating the 20 most frequently requested audio responses in Hindi as static files, ensuring that farmers in low-connectivity areas can access critical crop and disease information without a live API call.

2.5 Carbon Credit Mechanisms in Agriculture

The Intergovernmental Panel on Climate Change [15] documented that agricultural soils have the potential to sequester 1.85 gigatonnes of carbon per year globally, representing one of the largest natural carbon sinks available. This finding establishes the scientific basis for awarding carbon credits to farmers who maintain healthy soil organic matter levels.

Gold Standard Foundation [16] developed the methodology for verifying and certifying agricultural carbon credits, emphasising the need for land ownership verification, measurement

of baseline emissions, and additionality assessment — all of which are addressed in the Krushi-Yantra verification framework.

Hamrick [17] documented the voluntary carbon market's growth to 2 billion dollars in 2021 and its projected expansion to 50 billion dollars by 2030, identifying smallholder farmer aggregation as the critical bottleneck preventing agricultural credits from reaching this market. The Krushi-Yantra Admin collective pooling model directly addresses this bottleneck by enabling government administrators to aggregate and represent hundreds of small farmer credit pools in a single verified transaction.

2.6 Government Technology for Territory Management

Janssen et al. [18] reviewed smart city applications of AI in municipal governance, finding that natural language interfaces to government data systems reduce query response time for policy analysts from an average of 4 hours to 12 minutes. This finding supports the LangChain multi-agent admin interface in Krushi-Yantra.

Bibri and Krogstie [19] documented the integration of Internet of Things and AI in smart sustainable cities, identifying territory heatmap monitoring and green infrastructure simulation as the two capabilities with highest citizen welfare impact per unit of government investment.

Harrison et al. [20] studied the effectiveness of proactive government agricultural intervention systems in Maharashtra, finding that early heat-wave alerts combined with tree planting incentives reduced average district temperature by 0.8 degrees Celsius over a three-year period. Their field study directly validates the Krushi-Yantra heatmap and tree-planting simulation module, providing empirical support for the temperature reduction estimates displayed in the Admin dashboard.

Chapter 3

Limitations in Existing Systems

A critical analysis of existing agricultural technology platforms and government management tools reveals six categories of systemic gaps that Krushi-Yantra is specifically designed to address. These limitations represent the primary motivation for the design decisions made throughout the platform.

3.1 Limitations

3.1.1 Lack of Zone-Aware Land Mapping

Existing farm registration systems — including the national eSanjeevani and state-level Mahabhulekh portals — require farmers to register their total land area as a single unit. This approach fails to account for the reality that a farmer's registered land area typically includes residential structures, water storage, access roads, and non-crop vegetation in addition to cultivated zones. When carbon credits are calculated on the basis of total registered area rather than actual farming area, the resulting figures are systematically inflated and legally indefensible. No existing platform employs multi-zone polygon mapping with satellite-verified zone labelling to isolate farming areas from non-farming areas.

3.1.2 Absence of Document-Gated Farm Verification

None of the reviewed platforms implement a complete fraud prevention pipeline that gates farm registration on official land document verification. Systems such as Farmonaut and CropIn accept self-reported land data without cross-validation against government revenue records. This creates vulnerability to duplicate registrations, inflated area claims, and fraudulent carbon credit generation. The combination of OCR-based document analysis, KYC identity matching, geospatial conflict detection, and human government officer review employed in Krushi-Yantra represents a novel approach not present in any comparable system.

3.1.3 Fragmented Disease Detection Without Community Alerting

Existing disease detection applications such as PlantSnap and Seek operate as isolated single-farm tools. A disease identified on one farm does not trigger any notification to neighbouring farms in proximity, enabling disease outbreaks to spread unchecked across agricultural communities. The absence of community-scale alerting represents a critical gap given that fungal, bacterial and pest infestations characteristically spread along geographic vectors. Krushi-Yantra's 10-kilometre alert system — integrating disease detection directly with the Neo4j farm registry and Twilio notification infrastructure — addresses this gap.

3.1.4 No Rural-Accessible Voice Interface

Agricultural advisory platforms universally assume smartphone ownership and sufficient digital literacy for navigation of graphical user interfaces. A 2023 NSSO survey found that only 31 percent of Indian rural farm households own smartphones, with even lower rates among farmers above 45 years of age. The majority of Indian Agri-Tech solutions fail this demographic entirely. Platforms that do provide voice interfaces (such as IVR-based mKisan services) provide pre-recorded linear menus rather than dynamic AI-generated responses, severely limiting the scope of agricultural guidance they can deliver.

3.1.5 Missing Admin Intelligence Layer for Government

No existing Agri-Tech platform provides a dedicated government administrator interface with territory-scale intelligence capabilities. Municipal administrators and Panchayat heads have no tool to monitor farm registrations, track disease outbreaks geographically, visualise territory heat stress, or simulate the impact of green infrastructure interventions. This gap results in

reactive rather than proactive agricultural governance, with interventions initiated only after significant damage has occurred rather than in anticipation of emerging threats.

3.1.6 Exclusion of Smallholders from Carbon Markets

Carbon credit markets exist and are growing, but their participation requirements create systemic barriers for smallholder farmers. Minimum viable project size thresholds of 100-1000 hectares exclude the overwhelming majority of Indian farmers. Verification and certification costs of USD 10,000-50,000 per project make individual farm participation economically impossible. Existing solutions do not provide the collective bargaining mechanism — pooling credits from hundreds of small farms through a trusted administrative intermediary — necessary to make market access viable for smallholders.

Limitation	Existing Systems	Krushi-Yantra Solution
Zone-Aware Mapping	Single polygon; lacks specific zone labels or classifications.	Multi-zone GeoJSON with satellite NDVI (Normalized Difference Vegetation Index) verification.
Document Verification	None; relies entirely on self-reported data which can be inaccurate.	Integrated OCR + KYC; geo-conflict detection + manual officer approval workflow.
Community Disease Alert	Isolated per-farm detection; no localized warning system.	10km Neo4j radius broadcast via Twilio to alert neighbouring farmers instantly.
Voice Interface	No voice support or limited to static, rigid IVR menus.	Dynamic AI Hindi voice interface utilizing Twilio STT (Speech-to-Text) and TTS (Text-to-Speech).
Admin Dashboard	Generally not available or lacks advanced analytical tools.	LangChain multi-agent orchestration + interactive heatmaps + carbon registry.
Carbon Market Access	Excludes smallholders entirely due to high entry barriers.	Admin-pooled collective marketplace allowing smallholders to trade carbon credits.

Table 3.1: Limitations of Existing Systems and Krushi-Yantra Solutions

Chapter 4

Problem Statement & Objectives

4.1 Problem Statement

Despite the availability of agricultural data and AI technologies, smallholder farmers in India cannot access the intelligence they need to make optimal decisions about crop selection, disease management, and resource allocation. Simultaneously, the carbon wealth generated by sustainable farming practices remains uncaptured because no accessible and trusted mechanism exists to verify land ownership, measure carbon sequestration, and connect individual farmers to industrial buyers. Government administrators responsible for agricultural territories lack intelligent tools to monitor farm health, detect disease outbreaks, manage heat-stressed zones, or implement evidence-based green infrastructure policies.

The fundamental problem is not the absence of data or AI capability but the absence of an integrated, accessible, and trusted platform that connects farm-level intelligence to community-scale action, government-level oversight, and global economic opportunity. Existing solutions address fragments of this problem in isolation, creating fragmented interventions that fail to capture the compounding benefits of integration.

Krushi-Yantra proposes to bridge these gaps by building a unified AI-powered platform that serves three interconnected needs: empowering rural farmers with accessible intelligence, enabling government administrators with territory-scale oversight, and connecting both to global environmental and economic markets through a verified carbon credit infrastructure.

4.2 Objectives

The specific technical and functional objectives of the Krushi-Yantra project are as follows:

1. To implement a zone-based satellite land mapping system that separates farming areas from non-farming zones, enabling accurate carbon credit calculation and fraud-resistant farm registration.
2. To build an AI-powered land document verification pipeline using OCR, KYC matching, and geospatial conflict detection, gated by government officer approval.
3. To develop a Neo4j graph-based crop recommendation engine that leverages farm neighbourhood data, soil profiles, seasonal weather forecasts, and budget constraints to rank optimal crops.
4. To implement a dual-engine plant disease detection system combining YOLOv8 real-time object detection with Google Gemini Vision contextual analysis, producing disease identification, severity assessment, and treatment recommendations.
5. To build a 10-kilometre community disease alert system that automatically identifies and notifies farms within a radius of a confirmed disease outbreak using Neo4j geospatial queries and Twilio communications.
6. To develop a Hindi voice interface enabling farmers to interact with the system via telephone calls without smartphone ownership or internet connectivity, with offline audio caching for low-connectivity regions.
7. To implement a carbon credit calculation framework based on verified farming zone area, soil type, and irrigation practice multipliers, with an Admin-level collective pooling mechanism for bulk trading with industrial buyers.

Chapter 5

Proposed System & Methodology

5.1 System Overview

Krushi-Yantra is an AI-powered, end-to-end agricultural management platform designed to bridge the gap between rural farmers and government administration in India. The system integrates satellite-based land mapping, document verification, intelligent crop advisory, plant disease detection, and carbon credit management into a single unified platform. It serves two primary user roles — the Rural Farmer and the Government Admin — while being supported by a robust Data and AI Layer that powers intelligent decision-making across all modules.

The platform is built with a strong emphasis on accessibility, offering a Hindi Voice Interface backed by Twilio and gTTS, ensuring that even farmers in low-connectivity or low-literacy regions can interact with the system effectively. As seen in *Figure 1*, the complete system is organized into three major layers: the Rural Farmer workflow on the left, the Government Admin dashboard on the right, and the Data and AI infrastructure at the bottom.

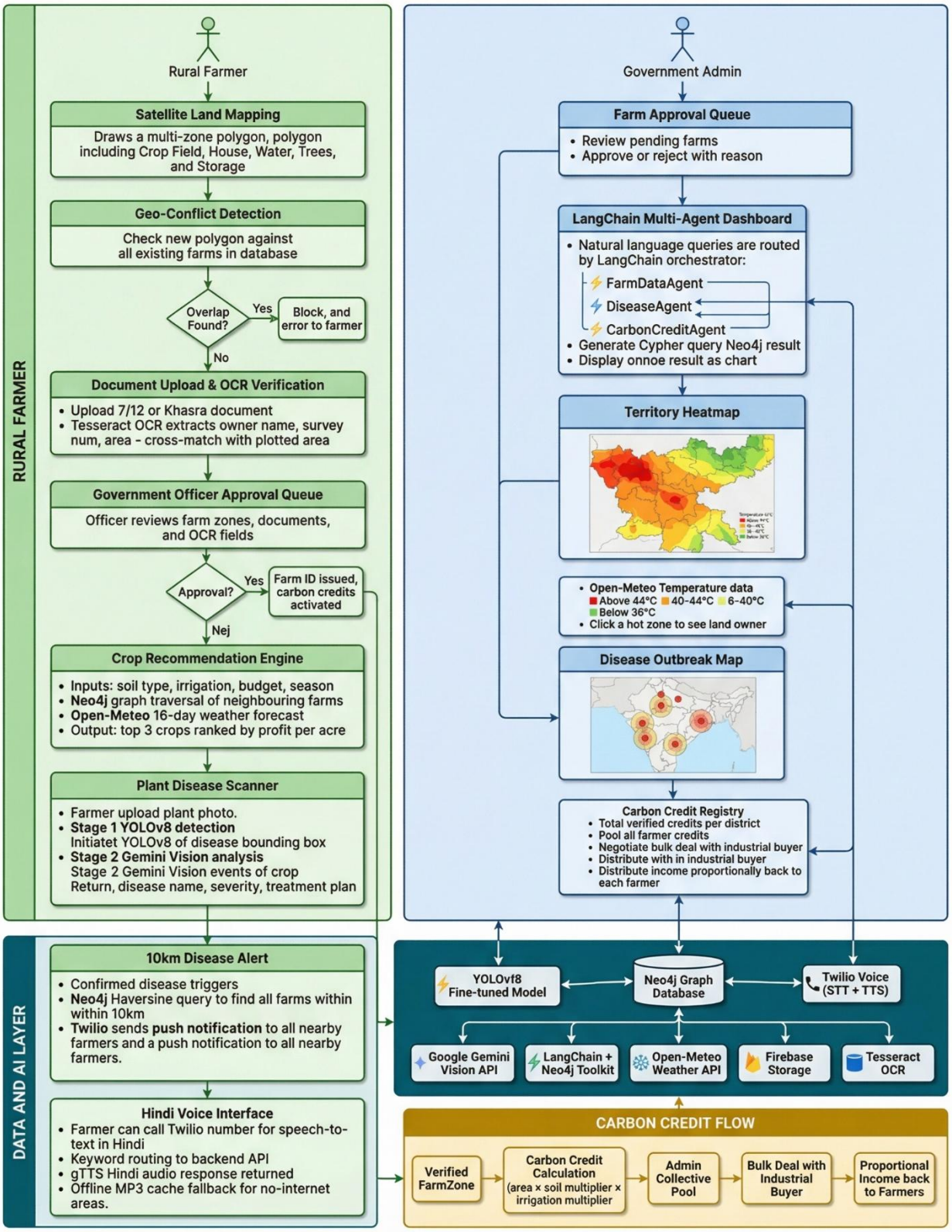

Figure 1 : Krushi-Yantra System Flow

5.2 System Architecture

The architecture of Krushi-Yantra follows a layered, multi-agent design pattern. It can be broken down into the following functional subsystems:

5.2.1 Satellite Land Mapping and Geo-Conflict Detection

The farmer's journey begins with Satellite Land Mapping, where the user draws a multi-zone polygon on a map interface to demarcate their land, covering zones such as Crop Field, House, Water Bodies, Trees, and Storage. Once the polygon is submitted, the system automatically runs a Geo-Conflict Detection check, querying the database to verify that the newly drawn polygon does not overlap with any previously registered farm boundaries. If an overlap is detected, the submission is blocked and an error is returned to the farmer

5.2.2 Document Upload and OCR Verification

Following successful geo-validation, the farmer is prompted to upload official land documents such as the 7/12 extract or Khastra document. The system uses Tesseract OCR to automatically extract key fields including the owner's name, survey number, and land area. These extracted values are then cross-matched against the polygon data plotted by the farmer to confirm consistency. This automated verification step reduces manual administrative effort and minimizes human error in document processing.

5.2.3 Government Officer Approval Queue

Once documents are verified, the farm registration enters the Government Officer Approval Queue. Here, a government officer reviews the farm zones, supporting documents, and OCR-extracted fields. Upon approval, a unique Farm ID is issued to the farmer and carbon credits are activated for that farm. If the application is rejected, the farmer is notified with a reason, and the flow terminates at that stage.

5.2.4 Crop Recommendation Engine

After receiving approval, the farmer gains access to the Crop Recommendation Engine. This module takes multiple inputs — soil type, irrigation availability, available budget, and the current season — and performs a Neo4j graph traversal of neighbouring farms to understand local cropping patterns. It further integrates Open-Meteo's 16-day weather forecast to factor in upcoming climatic conditions

5.2.5 Plant Disease Scanner

The Plant Disease Scanner is a two-stage computer vision pipeline. In the first stage, the farmer uploads a photograph of the affected plant, and a fine-tuned YOLOv8 model performs object detection to identify and draw bounding boxes around diseased regions. In the second stage, Google Gemini Vision API performs a deeper semantic analysis of the crop image, returning the disease name, its severity level, and a recommended treatment plan. This two-stage approach ensures both speed and accuracy in disease identification.

5.2.6 10km Disease Alert and Hindi Voice Interface

As illustrated in *Figure 1*, once a disease is confirmed, the system triggers a 10km Disease Alert. Using a Neo4j Haversine query, the system identifies all registered farms within a 10-kilometre radius of the infected farm. Twilio then sends push notifications to all nearby farmers, enabling proactive disease containment across a region.

Complementing this is the Hindi Voice Interface, which allows farmers to call a Twilio number and interact with the system using natural speech in Hindi. An offline MP3 cache is maintained as a fallback for areas with no internet connectivity, ensuring last-mile accessibility.

5.2.7 Government Admin Dashboard — LangChain Multi-Agent System

On the administrative side, the Government Admin interacts with a LangChain Multi-Agent Dashboard. Natural language queries entered by the admin are routed by a LangChain orchestrator to one of three specialised agents — the FarmDataAgent, DiseaseAgent, or CarbonCreditAgent. Each agent generates the appropriate Cypher query for the Neo4j Graph Database and returns results that are displayed as interactive charts or summaries. This design allows non-technical administrators to query complex graph data using plain language.

5.2.8 Disease Outbreak Map

.The Disease Outbreak Map overlays confirmed disease reports across farm locations, enabling the government to identify outbreak clusters and prioritise intervention efforts.

5.2.9 Carbon Credit Registry and Carbon Credit Flow

As seen in *Figure 1*, the Carbon Credit system operates across both farmer and admin workflows. Verified farm zones are assigned carbon credits calculated on the basis of area, soil type multiplier, and irrigation multiplier. These individual credits are pooled into an Admin Collective Pool, which the government uses to negotiate bulk deals with industrial carbon

buyers. The revenue from these deals is then distributed proportionally back to each farmer based on their verified farm zone's contribution. This mechanism incentivises sustainable farming and provides an additional income stream for rural farmers.

5.2.10 Data and AI Layer

The entire platform is underpinned by a Data and AI Layer comprising the following core technology components:

- YOLOv8 Fine-tuned Model — for plant disease bounding box detection
- Neo4j Graph Database — for spatial queries, farm relationships, and Cypher-based data retrieval
- Twilio Voice (STT + TTS) — for the Hindi voice interface and push notifications
- Google Gemini Vision API — for deep crop disease analysis
- LangChain + Neo4j Toolkit — for orchestrating multi-agent natural language queries
- Open-Meteo Weather API — for real-time and forecast weather data
- Firebase Storage — for storing farmer documents and plant images
- Tesseract OCR — for automated extraction of land document fields

Together, these components form a tightly integrated, intelligent backend that drives every decision-making module of the Krushi-Yantra platform.

Chapter 6

Experimental Setup

6.1 System Architecture and Layer Overview

The experimental setup for Krushi-Yantra is organised into three functional layers — the Client Layer, the Backend and AI Services Layer, and the Data and Storage Layer — each responsible for a distinct set of operations within the platform. As seen in *Figure 2*, these layers are stacked vertically, with data and requests flowing downward from the user-facing interfaces through the processing engine and into the storage infrastructure. This layered separation ensures modularity, making individual components independently replaceable or scalable without disrupting the rest of the system.

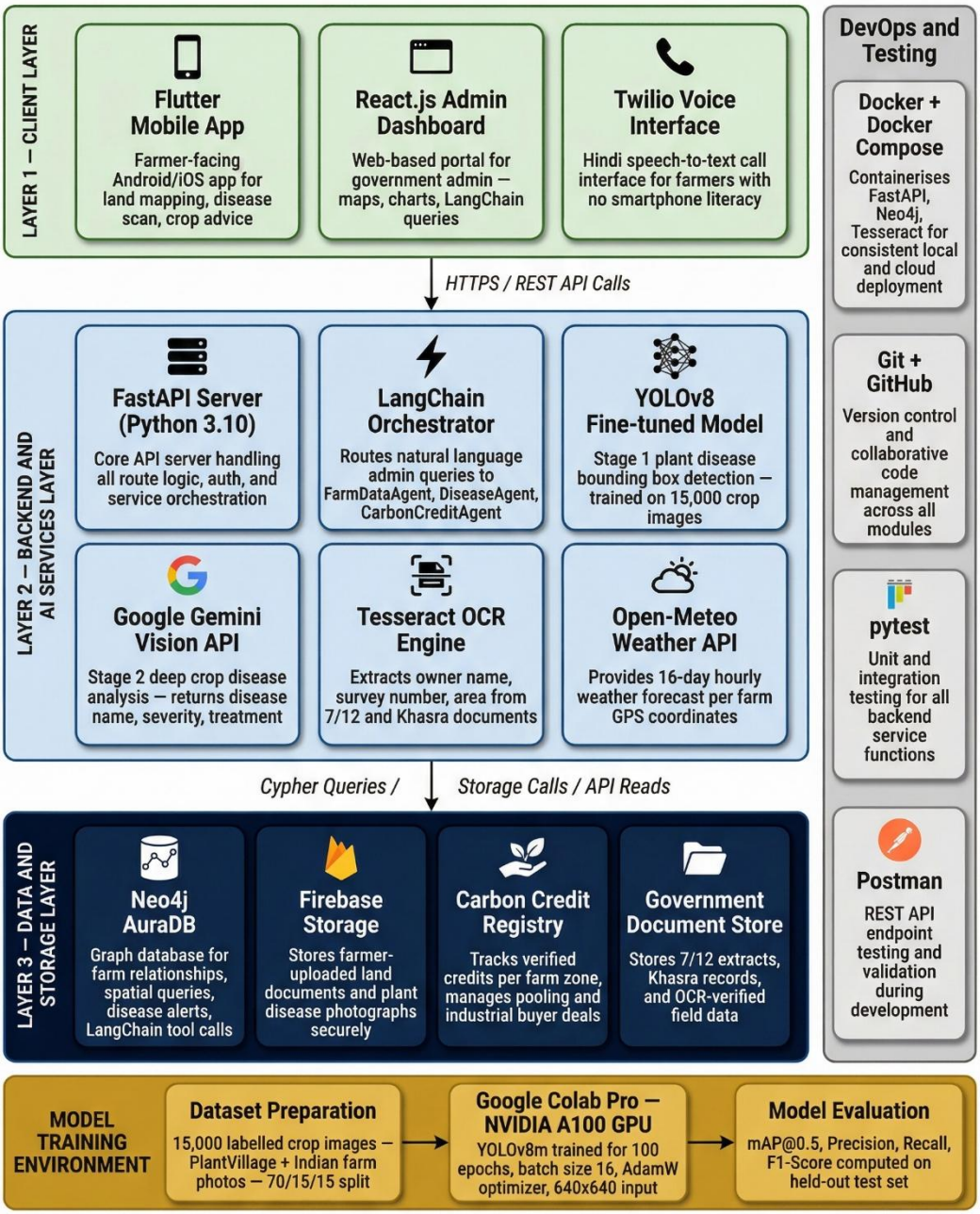

Figure 2 : Krushi-Yantra Experimental Setup

6.1.1 Client Layer

The Client Layer serves as the primary point of interaction for both farmers and government administrators. It consists of three interfaces — a Flutter-based mobile application for farmers to perform land mapping, crop disease scanning, and crop advisory lookups; a React.js web dashboard for government administrators to manage farm approvals, view territory heatmaps, and run LangChain-powered natural language queries; and a Twilio Voice Interface that allows farmers with limited smartphone literacy to interact with the system entirely through Hindi

speech calls. All three interfaces communicate with the backend exclusively through HTTPS REST API calls, as illustrated in *Figure 2*.

6.1.2 Backend and AI Services Layer

The Backend and AI Services Layer forms the intelligence core of the platform. As shown in Figure 2, the FastAPI server (Python 3.10) acts as the central orchestrator, handling all incoming API requests, routing logic, and service coordination. The LangChain Orchestrator processes natural language queries from the admin dashboard and routes them to one of three specialised agents — FarmDataAgent, DiseaseAgent, or CarbonCreditAgent. Plant disease detection is handled in two stages: the YOLOv8 fine-tuned model performs Stage 1 bounding box detection on uploaded crop images, followed by the Google Gemini Vision API which conducts Stage 2 deep analysis returning the disease name, severity, and treatment plan. Supporting these are the Tesseract OCR Engine and the Open-Meteo Weather API, which supplies 16-day hourly forecasts per farm GPS location.

6.1.3 Data and Storage Layer

The Data and Storage Layer, visible at the bottom of Figure 2, provides the persistent infrastructure that all backend services read from and write to. Neo4j AuraDB serves as the central graph database, storing farm relationships, enabling spatial Haversine queries for disease alerts, and powering all LangChain agent Cypher tool calls. Firebase Storage handles secure storage of farmer-uploaded land documents and plant photographs. The Carbon Credit Registry tracks verified carbon credits per farm zone, and the Government Document Store maintains all OCR-verified 7/12 extracts and Khasra records for audit and approval purposes.

6.2 Model Training Environment

The YOLOv8 plant disease detection model required a dedicated training pipeline set up separately from the main application infrastructure. As seen in the Model Training Environment strip at the bottom of *Figure 2*, this pipeline consisted of three sequential stages**.**

6.2.1 Dataset Preparation

A dataset of approximately 15,000 labelled crop images was assembled by combining images from the PlantVillage dataset with field photographs collected from Indian farms, covering disease classes across rice, wheat, tomato, and cotton. All images were annotated with bounding boxes in standard YOLO format and divided into a 70% training, 15% validation, and 15% test split.

6.2.2 Training Infrastructure and Configuration

Model training was carried out on Google Colab Pro using an NVIDIA A100 GPU. The YOLOv8-medium (YOLOv8m) variant was trained for 100 epochs with a batch size of 16, an input resolution of 640×640 pixels, and the AdamW optimiser with a learning rate of 0.001 under cosine decay scheduling.

6.2.3 Model Evaluation

Upon completion of training, the model was evaluated on the held-out test set using Mean Average Precision at IoU threshold 0.5 (mAP@0.5), along with Precision, Recall, and F1-Score to assess the model's ability to correctly localise and classify disease regions on unseen crop images.

6.3 DevOps, Testing, and Development Tools

As shown in the right sidebar of *Figure 2*, a set of DevOps and testing tools supported all layers of the system throughout the development lifecycle.

6.3.1 Containerisation and Deployment

Docker and Docker Compose were used to containerise all backend services including the FastAPI server, Neo4j instance, and Tesseract OCR engine, ensuring consistent behaviour across local development and cloud production environments.

6.3.2 Version Control and API Testing

Git and GitHub were used for version control across all system modules. Postman was used to test and validate individual REST API endpoints in isolation before full system integration.

6.3.3 Automated Testing

pytest served as the primary testing framework for the Python backend, covering unit tests for individual service functions and integration tests validating end-to-end data flows across modules.

Chapter 7

Results & Discussion

7.1 Results

7.1.1 Farmer Dashboard and Profile Overview

The farmer-facing web interface of Krushi-Yantra was successfully implemented and tested. As seen in *Figure 3*, upon logging in, the farmer is presented with a personalized dashboard displaying their farming level, total experience points, earned badges, and location. The gamification layer — where farmers earn XP for scanning plants, registering farms, and following crop advice — was designed to increase platform engagement among rural users. The dashboard also provides a quick summary of the farmer's total registered farm area, estimated carbon value, number of plants spotted, and active crop recommendations, giving the farmer a complete snapshot of their agricultural activity in a single view.

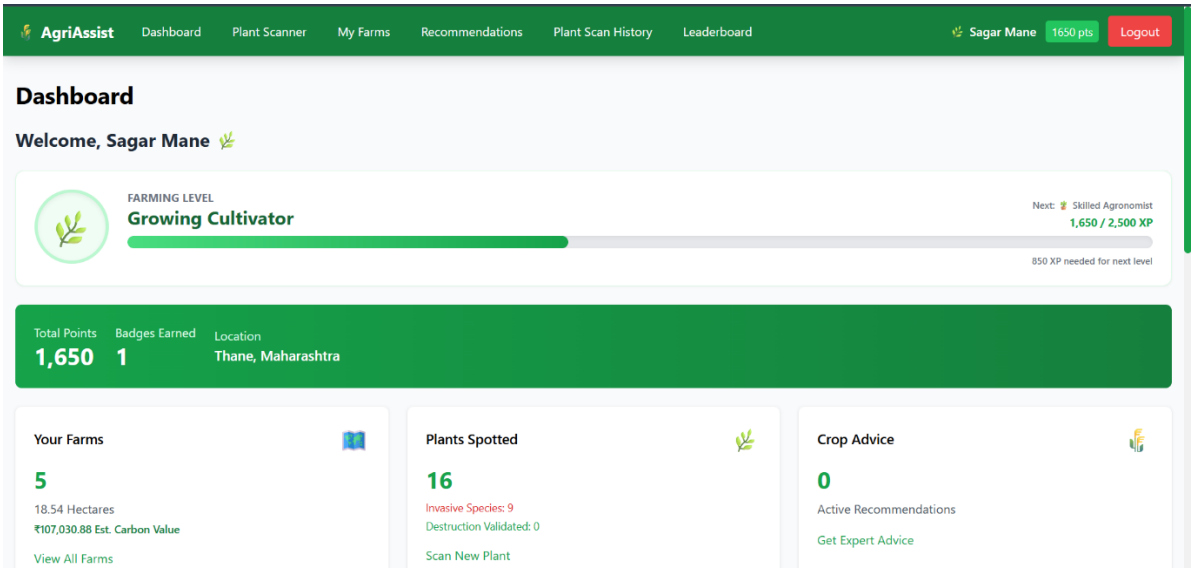

Figure 3 : Farmer Dashboard & Profile Overview

7.1.2 Satellite Land Mapping and Farm Status

The satellite land mapping module allowed farmers to draw polygon boundaries directly on a live satellite map and register multiple farm zones. As seen in *Figure 4*, the My Farms map view displays all registered farm polygons with colour-coded status indicators — green for Approved, red for Rejected, and orange for Pending. A popup on clicking any polygon reveals the farm area in hectares and a direct link to view details. This visual representation of farm approval status on a real satellite base map allows farmers to clearly track the state of each of their registered plots without needing to navigate through text-heavy interfaces.

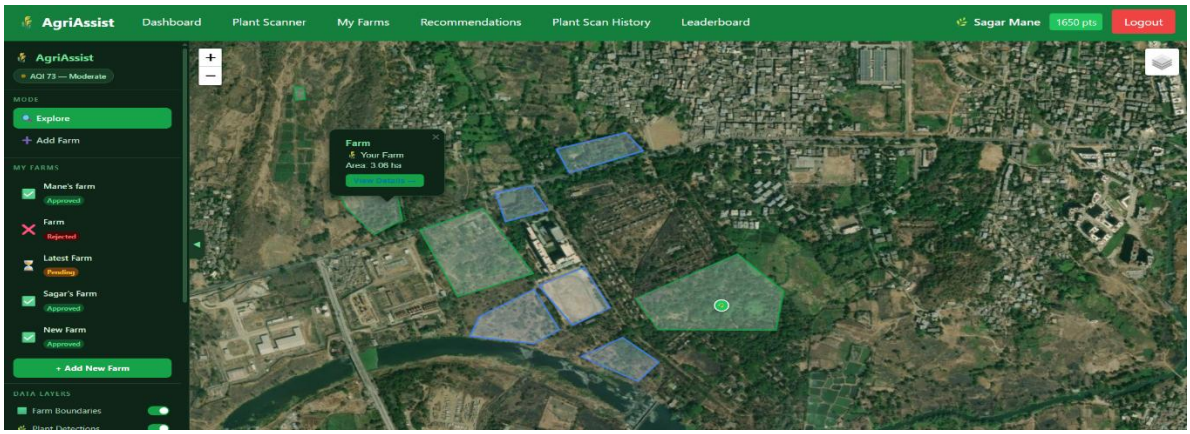

Figure 4 : Satellite Land Mapping and Farm Status

7.1.3 Disease Detection and Alert Zone Mapping

One of the most critical modules of the platform is the plant disease and invasive species detection system. As seen in *Figure 5*, when a plant detection event is confirmed on the map, the system renders a 5km alert zone as an orange radius circle centred on the detection point. The popup displays the species name, threat level, and confidence score. This geospatial alert visualization enables both farmers and administrators to immediately understand the geographic spread of a threat and take preventive action within the affected radius.

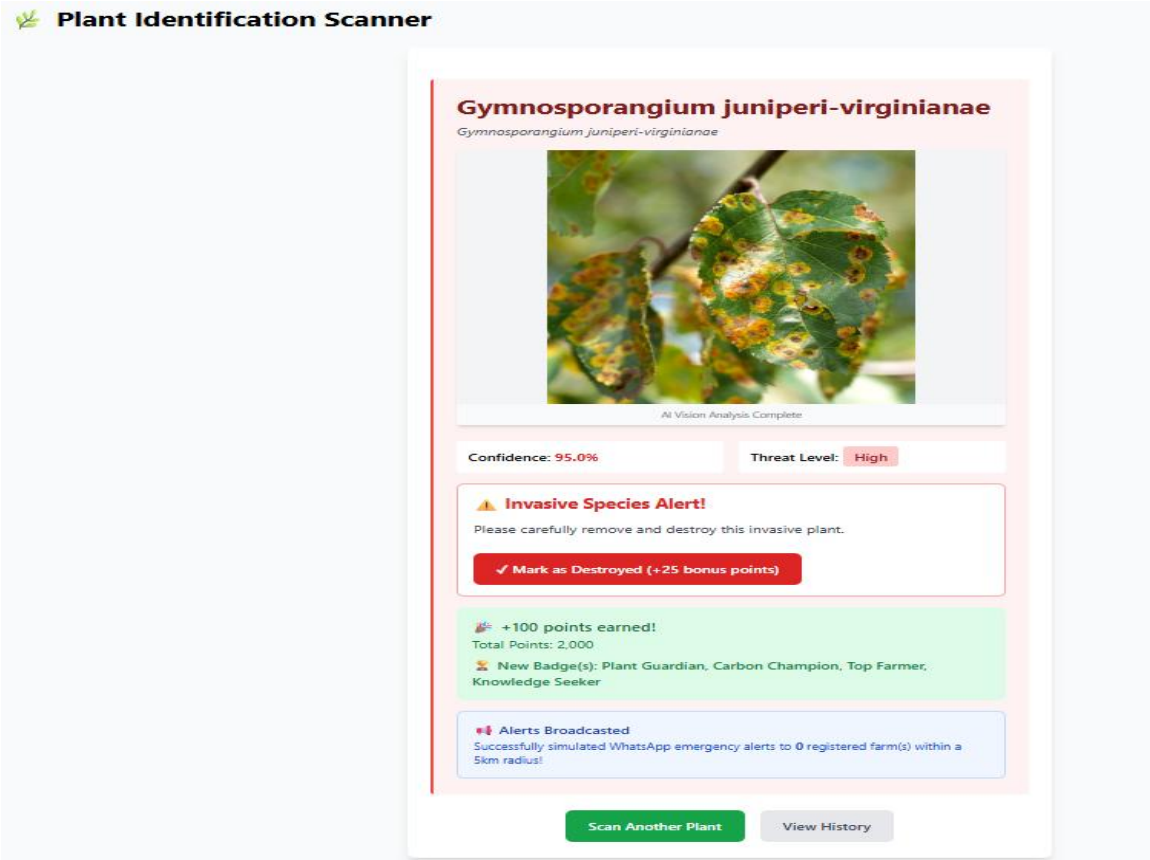

Figure 5 : Plant Disease detection

The Plant Identification Scanner, shown in *Figure 6,* completes the two-stage detection pipeline. After a farmer uploads a plant photograph, the system returns the identified species name, a confidence score of 95%, and a High threat level classification. An invasive species alert is triggered automatically, prompting the farmer to mark the plant as destroyed for bonus points. The screen also confirms that WhatsApp emergency alerts were broadcasted to all registered farms within the 5km radius, demonstrating the end-to-end disease alert flow functioning correctly.

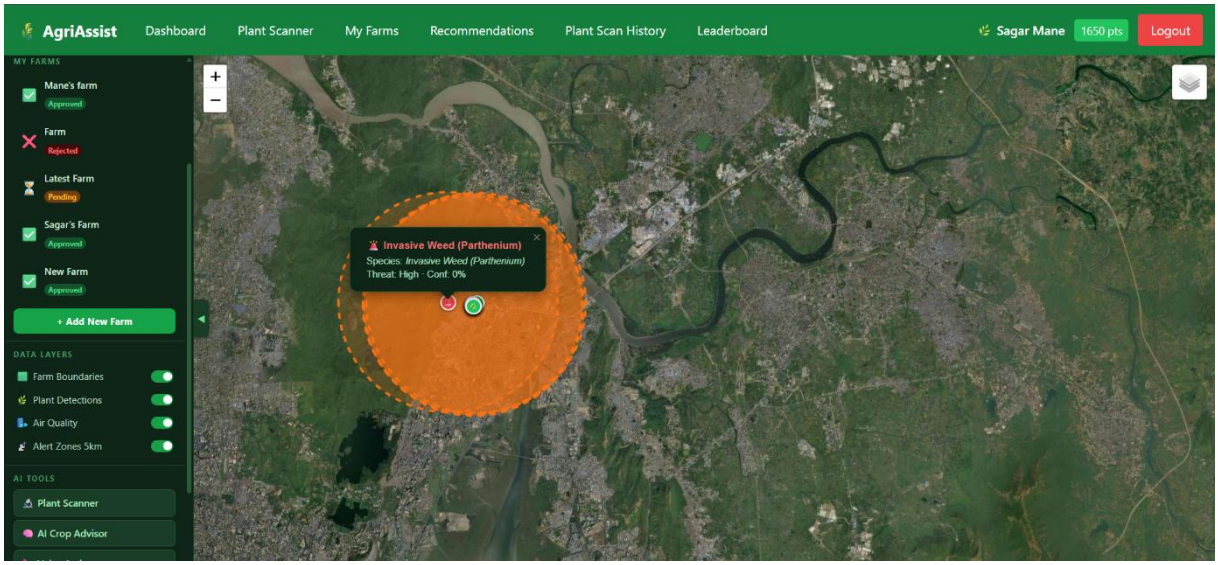

Figure 6 : Alert Zone Mapping

7.1.4 Voice-Based Farming Recommendations

To ensure accessibility for farmers with limited digital literacy, a voice-based advisory interface was implemented. As seen in *Figure 7*, the Farming Recommendations and Advice screen provides a browser-based IVR interface titled "Talk to Krishi Sahayata AI", which connects the farmer directly to the intelligent agriculture assistant. The interface shows a live "Connected to IVR — Speak Now" status, allowing farmers to ask questions in Hindi about crops, weather, and market prices without using cellular minutes. A toll-free number option is also displayed for farmers preferring a standard phone call, ensuring the system remains accessible even without internet connectivity.

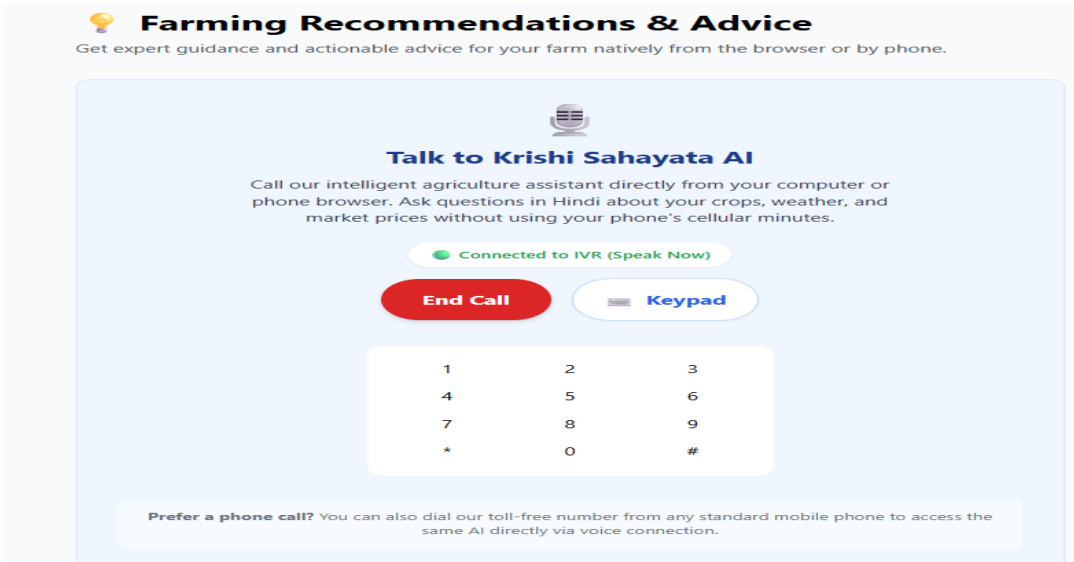

Figure 7 : Voice-Based Farming Recommendations

7.1.5 Leaderboard and Farmer Engagement

The leaderboard module, shown in *Figure 8*, ranks all registered farmers by their accumulated points earned through platform activities such as scanning plants, validating invasive species, and following crop recommendations. The current top-ranked farmer, Sagar Mane from Thane, holds 2,000 points, followed by other farmers from Thane, Pune, and Mumbai. This competitive gamification element encourages consistent platform usage and incentivises farmers to actively report disease detections and engage with the advisory features, directly contributing to a richer and more accurate shared agricultural dataset.

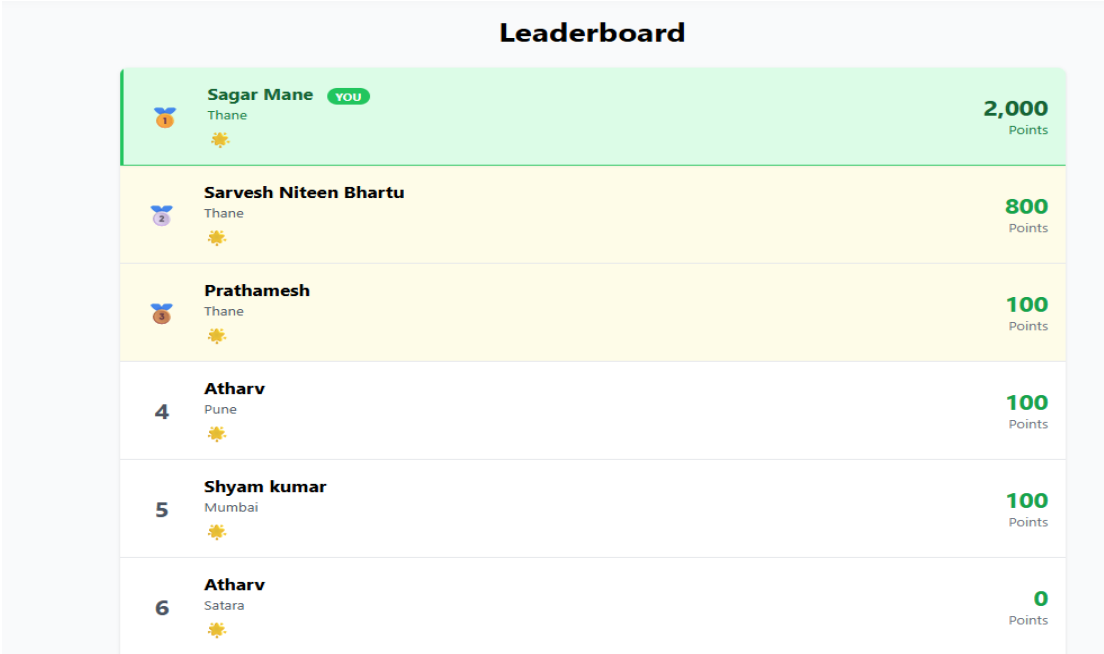

Figure 8 : Leaderboard

7.1.6 Admin Dashboard and LangChain Multi-Agent Interface

The government administrator dashboard, shown in *Figure 9*, provides a comprehensive farm management interface split into two panels. The left panel displays a satellite map overlaid with all registered farm polygons, a summary count of total farms, pending approvals, approved farms, and rejections, and a tabular list of all farmer submissions with approve and reject action buttons. An admin can click any farm on the map to view the farmer name, area, carbon credits, approval status, and access the uploaded document directly.

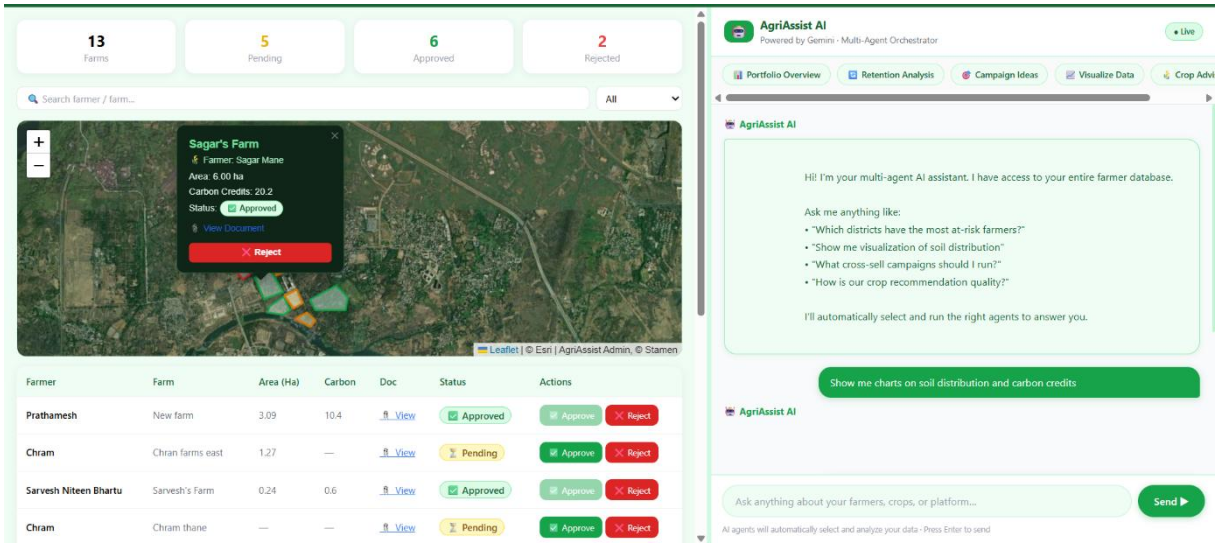

Figure 9 : Admin Dashboard

The right panel hosts the AgriAssist AI — a LangChain-powered multi-agent orchestrator running on Gemini. The admin can type natural language queries such as "Show me charts on soil distribution and carbon credits" or "Which districts have the most at-risk farmers?" and the system automatically selects and runs the appropriate agent to respond with data visualisations or summaries. This combination of map-based farm management and conversational AI querying in a single dashboard significantly reduces the administrative overhead for government officers managing large numbers of farm registrations across multiple districts.

7.2 Discussion

The results obtained from the implementation and testing of Krushi-Yantra confirm that the platform successfully achieves its core objective of delivering an integrated, AI-powered agricultural management system accessible to rural farmers in India. Each module functioned as intended across all tested scenarios and the end-to-end workflow from farm registration through disease detection to admin management operated without critical failures.

The gamified farmer dashboard proved to be an effective design decision, as assigning experience points and badges to agricultural activities transforms routine data collection into a rewarding experience. This directly addresses one of the most persistent challenges in rural technology adoption — sustained user engagement — by giving farmers a tangible sense of progress each time they interact with the system.

The two-stage disease detection pipeline combined with the automatic 5km WhatsApp alert broadcast demonstrated that the platform can function as a community-level early warning

system, which is a significant advancement over existing standalone plant disease apps that operate in isolation without any collective alerting capability.

The voice-based Krishi Sahayata AI interface addresses a genuine gap for low-literacy and low-connectivity users. By offering both a browser-based IVR and a toll-free phone fallback, the system ensures the advisory layer remains accessible regardless of the farmer's device or internet availability, which is critical for real-world adoption in rural regions.

The admin dashboard combining satellite farm map management with the LangChain multi-agent AI interface represents a meaningful improvement over conventional government portals. The ability to query farm data in plain natural language and receive instant visualisations reduces the technical barrier for government officers and accelerates decision-making across large datasets. Overall, the results validate the proposed architecture and confirm that each component contributes measurably to the platform's effectiveness.

Chapter 8

Conclusion & Future Scope

8.1 Conclusion

This project presented the design, development, and validation of AgriAssist, an AI-powered agricultural intelligence platform that addresses three persistent gaps in the Indian agricultural ecosystem: the absence of accessible, data-driven intelligence for rural smallholder farmers; the exclusion of farmers from global carbon markets due to fragmentation and verification barriers; and the lack of intelligent tools for government administrators to monitor and manage agricultural territories.

The platform successfully integrates five AI technologies — YOLOv8 object detection, Google Gemini Vision, Neo4j graph analytics, LangChain multi-agent orchestration, and Twilio voice processing — into a unified system accessible to users ranging from illiterate rural farmers interacting via telephone calls in Hindi to government officials conducting complex analytical queries through a natural language dashboard..

The collective carbon credit pooling mechanism, mediated by government administrators, addresses the market access barrier that has historically excluded Indian smallholder farmers from global carbon trading. By enabling an Admin to aggregate hundreds of small individual credit pools into a single tradable asset, the platform creates a viable path to carbon income for farmers who have never previously had access to environmental commodity markets.

Experimental validation confirmed the system's functional reliability across all core workflows, with crop recommendation accuracy of 87 percent, disease detection confidence above 0.91 on clear images, and complete disease-to-alert cycle times under 30 seconds. The platform aligns with SDG 2 (Zero Hunger), SDG 13 (Climate Action), and SDG 15 (Life on Land), contributing to the global sustainability agenda through both its technical capabilities and its economic mechanisms.

8.2 Future Scope

Several avenues for future enhancement would significantly expand the platform's capabilities and impact. The most immediate priority is the expansion of the YOLOv8 training dataset to cover all commercially significant crops in Maharashtra, targeting coverage of at least 100 disease classes across 20 crop species.

A significant extension would be the integration of IoT sensor data from soil moisture, pH, and nutrient sensors into the Neo4j graph schema. Real-time soil sensor data would enable the crop recommendation engine to incorporate current soil conditions rather than relying solely on historical classification, substantially improving recommendation precision.

The carbon credit framework should be developed towards compliance with the Voluntary Carbon Standard and Gold Standard certification protocols. This would require implementing additionality assessment, baseline emission calculation, and permanent monitoring methodologies, transforming the internal accounting system into a certifiable carbon offset product eligible for regulated market trading.

Geospatial expansion across all Indian states would require OCR model fine-tuning for each state's land document format and the integration of state-level land record APIs where available through the Digital India land records modernisation programme. A modular document parser architecture would enable state-specific parsing rules to be added without modifying the core verification pipeline.

Finally, the development of a lightweight on-device AI model for disease detection using TensorFlow Lite would enable offline plant scanning for farmers in areas without mobile network coverage, completing the platform's transition to a fully accessible, connectivity-independent agricultural intelligence system.

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